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The Palaeomagnetism of Peat's Ridge Dolerite and Mt. Tomah Basalt

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Abstract—The palaeomagnetism of basic igneous rocks at Peat's Ridge and Mount Tomah, N.S.W., have recently been reported by Manwaring (1963); after partial demagnetization in alternating magnetic fields the directions of magnetization of the specimens from Peat's Ridge were consistent but those from Mt. Tomah remained scattered. The author has studied specimens from the same occurrences using partial *thermal* demagnetization techniques and has obtained results closely comparable with those of Manwaring. The unsuitability of the Mount Tomah basalt for palaeomagnetic purposes is discussed briefly.

The palaeomagnetism of certain igneous rocks of the Sydney Basin was discussed by Manwaring (1963) and in particular reference was made to the fine grained olivine dolerite intrusion at Peat's Ridge (near Gosford, N.S.W.) and the basalt flows capping Mt. Tomah (in the Blue Mountains of N.S.W.). Manwaring calculated a palaeomagnetic pole position for the specimens obtained from Peat's Ridge and from this was able to make an estimate of the geological age of the intrusion. He was also able to show that the specimens so far collected from Mt. Tomah are unsuitable for palaeomagnetic purposes. Even after partial demagnetization in fairly high alternating magnetic fields there was very little reduction in the scatter of the magnetic vectors of these specimens.

The author has been engaged in a study of thermal demagnetization of material and in the course of this study laboratory-induced magnetic moments of igneous rock and " thermal washing " or partial thermal demagnetization experiments were carried out on a number of specimens from Peat's Ridge and Mt. Tomah. The purpose of this note is to compare the results of these experiments with those of Manwaring.

Methods Employed

In these experiments the specimens were progressively demagnetized by heating them in a non-inductive electric furnace, through which a supply of nitrogen was maintained, and then cooling them in magnetic field free space. The earth's magnetic field was cancelled by passing currents through a system of large Helmholtz-type coils. The maximum magnetic field that may be left uncompensated by the coils, without resulting in a thermo-remanent moment, is readily calculated by extrapolating the TRM data of Dickson (1962) to very low fields. For specimen T5C a value of 5×10^3 oersteds is indicated, which is substantially larger than the uncertainty in cancellation of the earth's field with the Helmholtz coils. However, as an additional precaution against the influence of stray laboratory fields, the specimens were rotated continuously about a vertical axis as they cooled; it was thus impossible for a stray horizontal field to have any effect unless it fluctuated coherently with the rotation of the specimens. The magnitude and direction of magnetization of each specimen was measured before and after each individual heating with an astatic magnetometer described by S. A. A. Kazmi (unpublished M.Sc. Thesis, University of Sydney, 1960). It was thus possible to remove the unstable magnetization and leave only the original thermo-remanent magnetization of the rocks.

Peat's Ridge

The NRM of a total of 20 specimens from five oriented samples collected by Manwaring was measured and then thermally washed in late 1961; all specimens were found to be reversely magnetized. The relative change in the intensity of magnetization, M_T/M_0, with rise in temperature is shown in Fig. 1 for a typical specimen from all but one sample; the curves obtained from the specimens of the remaining sample are very similar to those of Fig. 1 and only tend to confuse the diagram. Curve 3 is of particular interest as the initial rise in M_T/M_0 is due to the removal of a relatively unstable component of magnetization. The directions of magnetization were statistically analysed using Fisher's treatment of dispersion on a sphere (Fisher, 1953) and k is an estimate of Fisher's precision parameter.

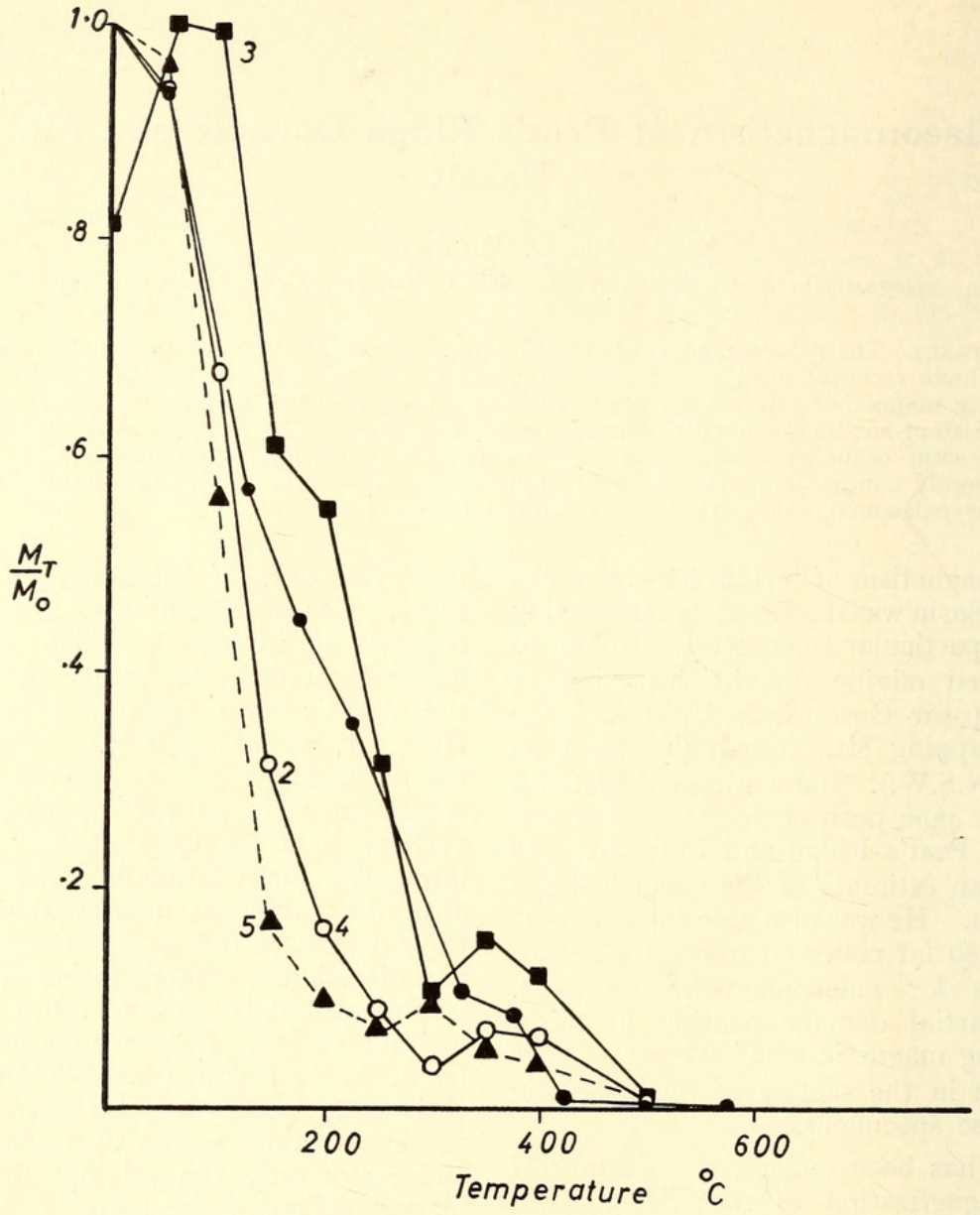


FIG. 1

The relative change in the intensity of magnetization with increase in temperature for the Peat's Ridge specimens. M_O is the value of the NRM and M_T is the value of the intensity of magnetization at temperature T

TABLE I

Average direction of magnetization and comparison between unwashed and thermally washed specimens from Peat's Ridge

S=number of samples ; N=number of specimens ; D=declination of average direction ; I=inclination of the average direction ; R_s=vector resultant of sample means ; R_N=vector resultant of specimens ; k_w=within-sample precision ; k_b=between-sample precision ; k_o=over-all precision ; a=half-angle of cone at 95% confidence (P=0·05) calculated from k_o

	S	N	D	I	R_s	R_N	k_w	k_b	k_o	a
NRM	5	20	176	+72	4·07	14·96	4	4	16	35°
Demag. 150°	5	20	193	+71	4·98	19·86	136	161	620	5·6°

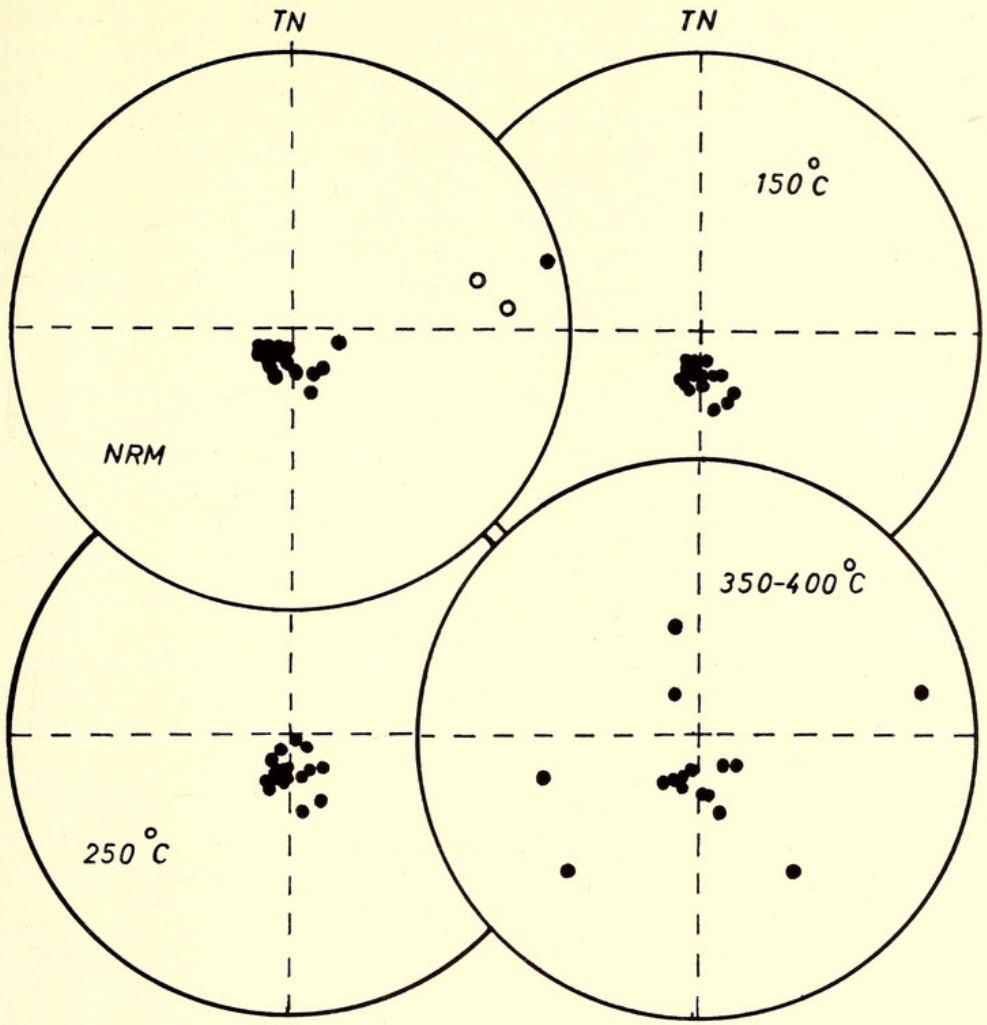


FIG. 2
Change in the direction of magnetization of the Peat's Ridge specimens with thermal washing

There is a considerable scatter in the NRM of the individual specimens ; so much so that even the within-sample precision is very low (k_w=4), Table 1. A calculation based on the data given for the NRM in Table 1 indicates that the half-angle of the cone of confidence at the 95% level is 35°. This is larger than that reported by Manwaring and is probably due to the fact that the specimens may have been exposed to stay laboratory fields during the early stages of the investigations. However, after washing the specimens at progressively higher temperatures the scatter of the magnetization vectors was reduced. The change in the direction of magnetization with rise in temperature is displayed in Fig. 2 and the best agreement was observed to occur at 150° C.

The results of the investigations made on Peat's Ridge have been presented in Table 1 in exactly the same form as that used by Manwaring to allow an easy comparison of the data to be made. However, if the procedure of taking only the sample mean directions is used, a slightly larger value (6°) is obtained for the half-angle at the cone of 95% confidence. This is due to the fact that there is a large difference in the number of specimens used within each sample ; this would also explain the fact that the within-sample precision is a little less than the between-sample precision after washing.

After thermal washing to 150° C the palaeomagnetic pole position calculated was 66° S, 134° E ; this compares very favourably with 70° S, 134° E, calculated by Manwaring. It is of further interest to note that the close agreement of results has been obtained by using entirely different methods of treatment ; alternating magnetic field " washing " on one hand and thermal " washing " on the other.

Mt. Tomah

In the middle of 1961 the effect of thermal demagnetization was observed on the NRM of 16 specimens taken from Mt. Tomah and it was

noted that "washing" did not reduce the scatter of the directions of magnetization, even at fairly high temperatures.

At the end of 1961 a fresh set of specimens, a total of 33 from three different sites, was collected from a new quarry face at Mt. Tomah. It was realized from previous experience that samples taken from the top ot Mt. Tomah would be unsuitable and it was thought that these later samples would be satisfactory, palaeomagnetically.

All 33 specimens were gradually demagnetized in the same way as before. The precision of specimens (two from each sample) chosen to represent their sampler, was initially very low, 2·3, and at no stage during demagnetization did this situation improve; even at 175° C the precision was only 2·5. On applying the randomness test given by Watson (1956) it is seen that the distribution of directions was random (at the 5% level of significance) both initially and at 175° C.

It is not understood why there should be such a scatter in the direction of magnetization; however it was noticed that one sample taken from a road cutting at the top of Mt. Tomah was very strongly magnetized, the average intensity of magnetization being $78{\cdot}5 \times 10^{-3}$ emu/cm^3, whereas the average for other samples taken from the same area was only $13{\cdot}5 \times 10^{-3}$ emu/cm^3. Experiments similar to those described by Dickson (1962) showed that the TRM produced by cooling the specimens, with the high value of NRM, in 1·0 oe. was much smaller than their NRM; and assuming that the earth's field was 0·6 oe. at the time when the original magnetization was induced, the present NRM is approximately five times the expected value. It seems likely, therefore, that the relatively high intensity of magnetization of this sample is due to lightning strike or possibly to some large man-made disturbances. The magnetization of the specimens from Mt. Tomah is held by a mineral phase with a fairly low Curie temperature (150°–200° C) and this has made it difficult to remove spurious effects without destroying the original magnetization. Also, as Manwaring suggested, the scatter in the direction of magnetization is probably due to the original thermo-remanent magnetization decaying with time leaving only randomly directed magnetizations.

Therefore it must be concluded, from the existing evidence, that the Mt. Tomah basalt is unsuitable for palaeomagnetic purposes.

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